

Morphology

- Morphology is the part of grammar that addresses the inner structure of words
- While English has a rather impoverished morphology, other languages have a much richer one (e.g., Hebrew)
- Morphological analysis is the decomposition of words into their sub-parts and the structure that combines them
- In NLP, morphological analysis minimally involves segmentation into tokens (the minimal units of analysis)
 - There are many tokenization schemes

Morphological Analysis in NLP

- Basic tokenization: (mostly separating out clitics)
 - In English, mostly involves separating out apostrophes:
 - isn't → is+n't
 - let's → let+'s
 - Dina's --> Dina+'s
 - ספר, ה', בית, ל' will turn into ['ל', 'בית', 'ה', 'ספר']
 - Auf'm → auf 'm (German: auf+dem)
- Lemmatization: mapping to the base form
 - dogs → dog
 - כלבים → כלב
 - כלב → كلب (lit. dogs → dog)
 - takes → take
 - נתן → יתנו

Segmentation into Morphemes

- Some languages combine words into compounds or other morphologically complex structures
 - *renewal* → *re* + *new* + *al*
 - *German: nachgedacht* → *nach* + *ge__t* + *dach* (lemma: *denken*)
 - *מטבחונים* → *ים* + *ון* + *מטבח*